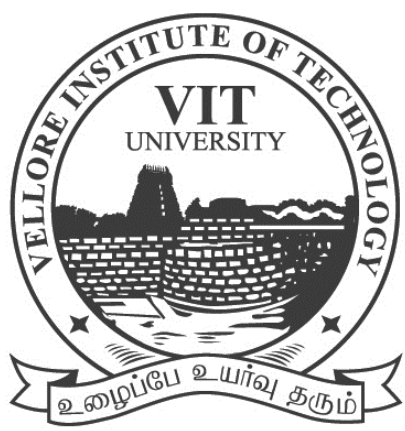




# Network Intrusion Detection System Using Reinforcement Learning and Automated Audit Reporting

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## Introduction

Traditional IDS rely on signatures and struggle with zero-day attacks. We use Reinforcement Learning (DQN & Actor-Critic) alongside ML to build an adaptive IDS that improves with time.

## Motivation

Cyberattacks like SQL injection and DDoS are becoming harder to detect. RL enables adaptive detection, improving resilience against evolving threats.

## Scope of the Project

The project builds an adaptive IDS by combining ML and RL models. It includes: Data preprocessing for clean, formatted input Supervised learning (Random Forest, Decision Tree) for known attacks .Unsupervised learning (K-Means) for anomaly detection. Reinforcement learning (DQN, Actor-Critic) for adaptability . The system supports real-time detection and is evaluated using accuracy, F1-score,  $R^2$ , and RMSE, demonstrating improved resilience against evolving threats.

## Methodology

**Data Preprocessing:** Scaling, encoding, train-test split

**Supervised:** RF, SVM, DT

**Unsupervised:** K-Means + PCA

**RL:** DQN, Actor-Critic with reward signals

**Testing:** CV, Grid Search, Confusion Matrix, ROC

## Results

**RF & LR:** >90% accuracy

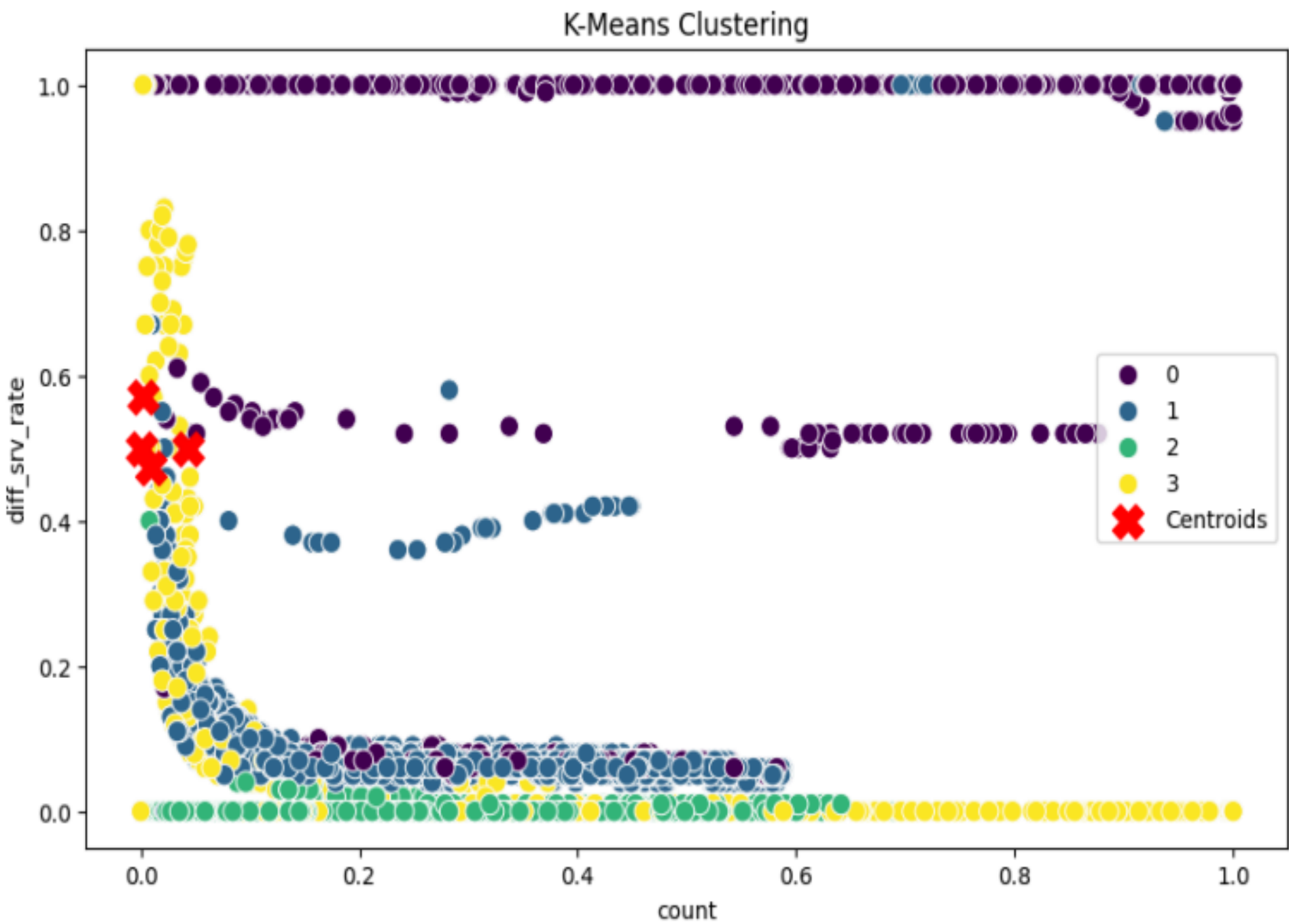
**K-Means:** Effective cluster separation

**DQN AUC**  $\approx$  0.93

**Actor-Critic:** Faster convergence

**RL** = better adaptability

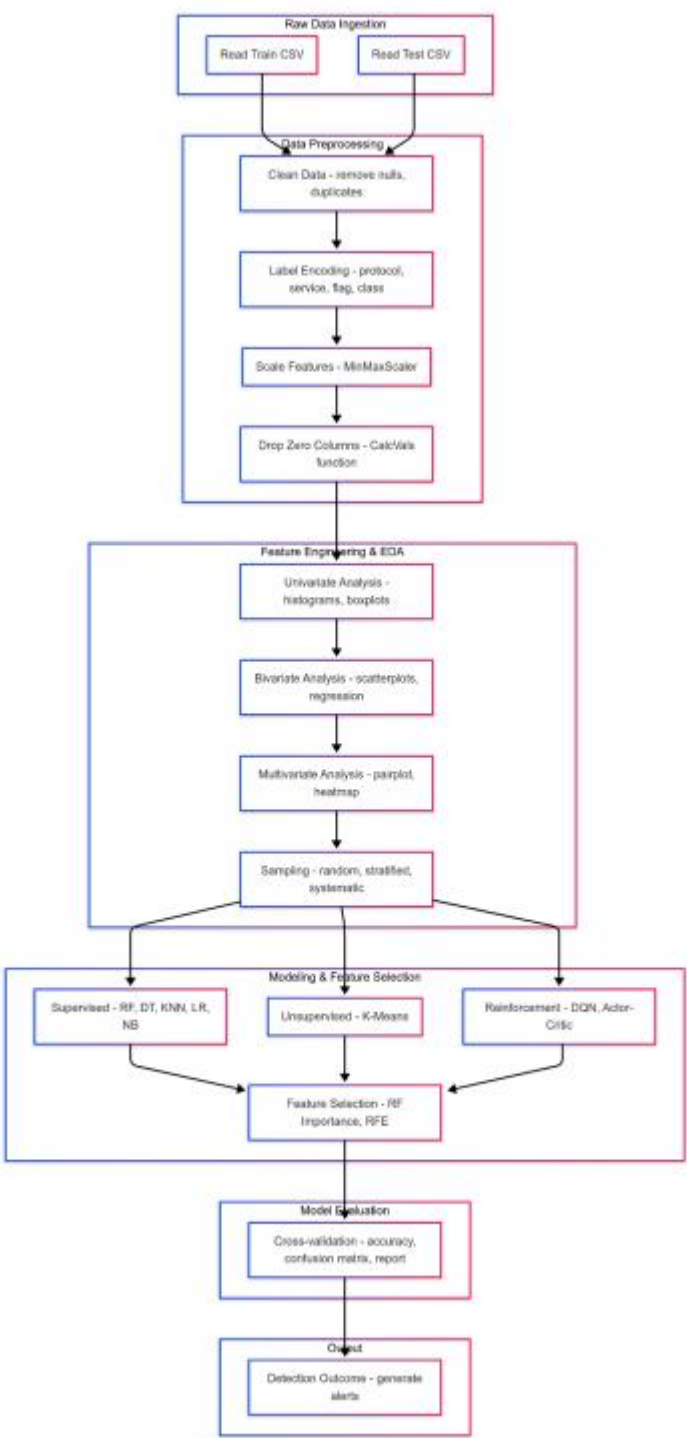
K Means Clustering Plot



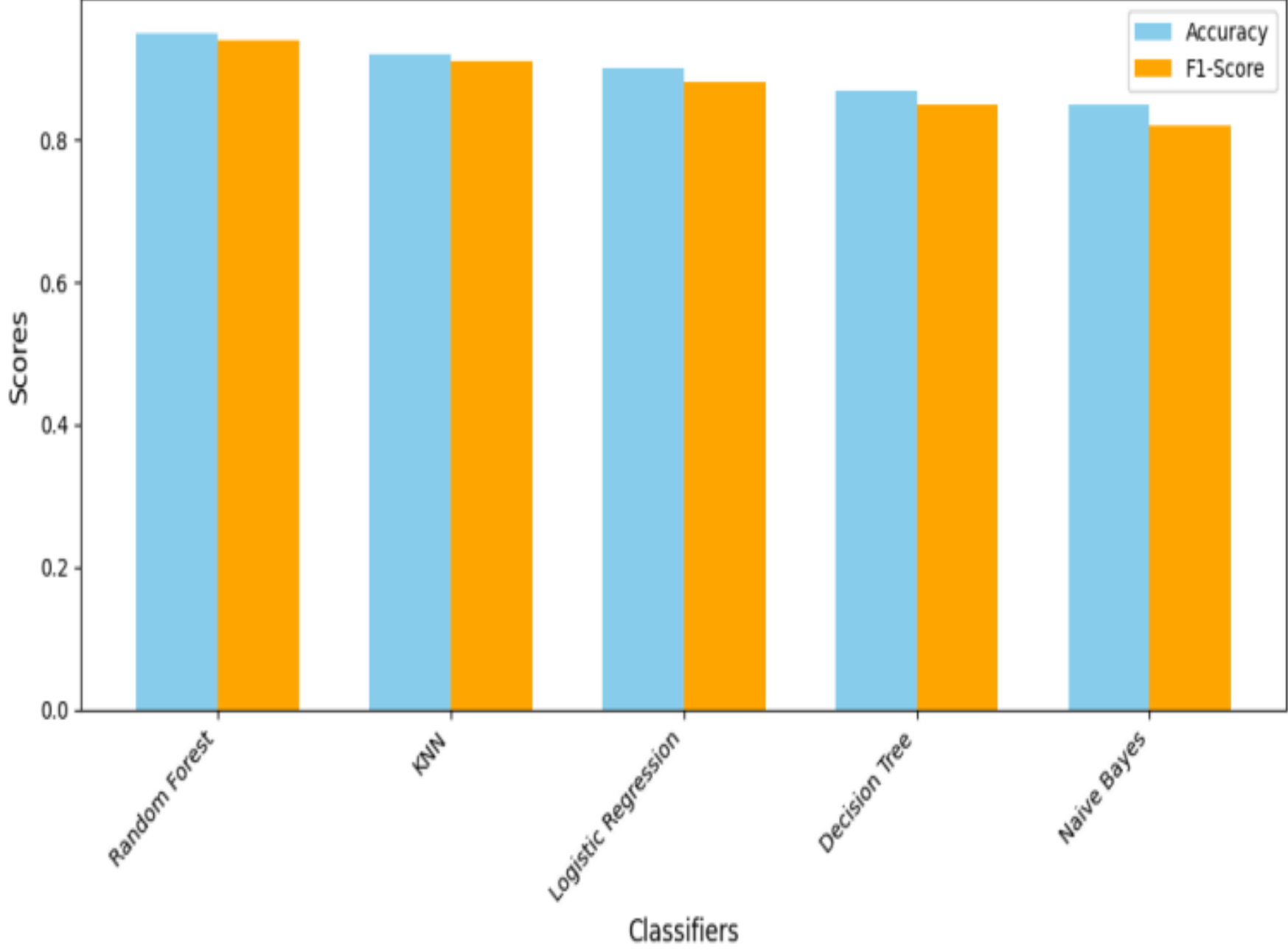
System Architecture



Data Flow Diagram



Performance Comparison of Classifiers (Figure 7.1)



## Conclusion

**Findings** : Hybrid ML methods improved intrusion detection and showed real-time potential.

**Hypothesis** : Supported – results confirmed better performance with reinforcement learning.

**Future Work** : Add deep learning, deploy with Docker, use explainable AI, and validate on more datasets.

## References

Mnih et al., 2015 – Deep RL for Human-level Control  
Scikit-learn Documentation  
CICDDoS2019 Dataset  
Kaggle SQL Injection Dataset